

ST5500

Hanging Transition Reader
for FX9600



ZEBRA

Installation Guide

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About This Document

Introduction

This guide provides instructions for mounting the FX9600 RFID reader to the ST5500 hanging transition reader and subsequently installing this assembled unit to the facility.



IMPORTANT: If you have a problem with your equipment, contact Zebra Global Customer Support for your region. Contact information is available at: www.zebra.com/support.

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Installation and servicing should be performed by qualified personnel, and all work should follow industry best practices and must conform to local codes.

Safety Precautions

- Potentially harmful voltages can be generated whenever the Appliance is connected to PoE. Therefore, avoid contact with electrically active parts internal to the Appliance.
- Only authorized and trained personnel should have access to or perform work on the Appliances.
- When working with the Appliances, remove all metallic jewelry, use properly insulated tools, and wear appropriate personal protective equipment (non-slip gloves, form fitting clothing, etc.) to reduce the risk of electric shock and mechanical entanglement.
- Only carry an Appliance by its frame with two or more people.
- Do NOT drop or place objects (such as tools) on the Appliances.
- Do NOT sit, stand, step on, jump or walk on the Appliances. Do NOT carry Appliances on your head.
- Do NOT damage or scratch the top, bottom, or side surfaces of the Appliances.
- Do NOT disassemble the Appliances or remove any part of the Appliances. The Appliance does not contain any user-serviceable parts.
- Do NOT allow the Appliances to get wet or expose the Appliances to direct sunlight for extended periods of time.
- Do NOT install Appliances near exposed flames, high heat sources, or flammable materials.

Unpacking and Storage

- At time of receipt, verify that the product delivered is in fact the product ordered. The product name, model number, and serial number of each Appliance are clearly marked on the outside of each packing box.
- Leave the Appliance in its original packing box until you are ready to install the Appliance.
- Store packing boxes in a clean, dry, ventilated area with relative humidity below 85% and ambient temperatures between -40°C and 70°C.
- Do NOT stack more than the maximum amount of allowable boxes on top of each other.
- Do NOT place Appliances directly on top of each other.
- At the installation site, keep Appliances, and in particular their electrical contacts, clean and dry before installation. If connector cables are left in damp or humid conditions then the contacts may corrode. Any cables with corroded contacts should not be used.
- If boxes are stored temporarily outside then place a protective covering over the boxes to protect them from direct weathering.
- Two people should unpack the Appliance from the packing box. When handling the Appliance always use both hands.
- Do NOT use a knife to cut the zip-ties, but instead, use wire cutting pliers.

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Chapter Descriptions

Topics covered in this guide are as follows:

- [Getting Started](#) provides information on the box contents, additional equipment, tools required, and product features.
- [Mounting FX9600 Reader to ST5500](#) provides information on mounting an FX9600 RFID reader to the ST5500.
- [Installing ST5500](#) provides information on ST5500 instructions and the installation options.

Notational Conventions

The following conventions are used in this document:

- **Bold** text is used to highlight the following:
 - Dialog box, window and screen names
 - Drop-down list and list box names
 - Check box and radio button names
 - Icons on a screen
 - Key names on a keypad
 - Button names on a screen.

- Bullets (•) indicate:
 - Action items
 - Lists of alternatives
 - Lists of required steps that are not necessarily sequential.
- Sequential lists (e.g., those that describe step-by-step procedures) appear as numbered lists.

Icon Conventions

The documentation set is designed to give the reader more visual clues. The following graphic icons are used throughout the documentation set. These icons and their associated meanings are described below.



NOTE: The text here indicates information that is supplemental for the user to know and that is not required to complete a task.



IMPORTANT: The text here indicates information that is important for the user to know.

Related Documents and Software

The following documents provide more information about the ST5500 Hanging Transition Reader.

- SmartLens™ for Retail Installation Reference Guide (p/n MN-002705-xx)

For the latest version of this guide and all guides, go to www.zebra.com/support.

Service Information

If you have a problem with your equipment, contact Zebra Global Customer Support for your region. Contact information is available at www.zebra.com/support.

When contacting support, please have the following information available:

- Serial number of the unit
- Model number or product name
- Software/firmware type or version number

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If your problem cannot be solved by Zebra Customer Support, you may need to return your equipment for servicing and will be given specific directions. Zebra is not responsible for any damages incurred during shipment if the approved shipping container is not used. Shipping the units improperly can possibly void the warranty.

If you purchased your Zebra business product from a Zebra business partner, contact that business partner for support.

Getting Started

Introduction

The ST5500 is a hanging transition reader that includes four antennas and four RF cables. Mount the FX9600 RFID reader (sold separately) to the ST5500 before installing the unit to the facility.

This chapter provides an overview on the box contents, additional equipment, tools required, and ST5500 features.

Box Contents

The ST5500 box contains the following:

- ST5500 Hanging Transition Reader - a single unit pre-assembled with four antennas and four RF cables
- Two carabiners
- Reader Installation kit (Poly Bag taped to the cross beam) with:
 - Four (4) 10-32, 5/16 lg screws to mount reader to mounting bracket
 - Two (2) ¼-20, 1.5 lg weight screws
 - Two (2) ¼-20 nuts
 - Six (6) flat washers, 1/2 in. ID (used as balancing weights).

Additional Equipment (Not Included)

- Surge protector (optional), Zebra p/n PWR-SGD00V30W-1NA
- Wire rope with a minimum diameter of 1/8 in. and a minimum load capacity of 200 lbs
- Wire rope clamps
- Cable ties
- Strut channel and mounting hardware (if required).

Tools Required

- Wire rope cutter
- 5/16 in. nut driver
- Level
- Drill (for drywall or drop ceiling installations).

ST5500 Features

Figure 1 Isometric View

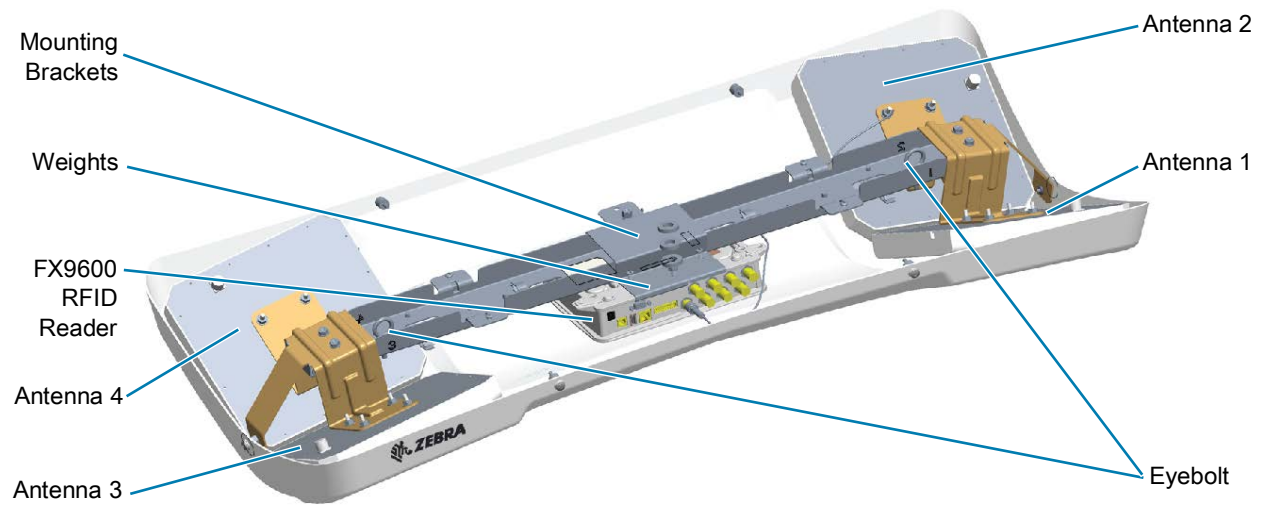


Figure 2 Front View



Figure 3 Top View

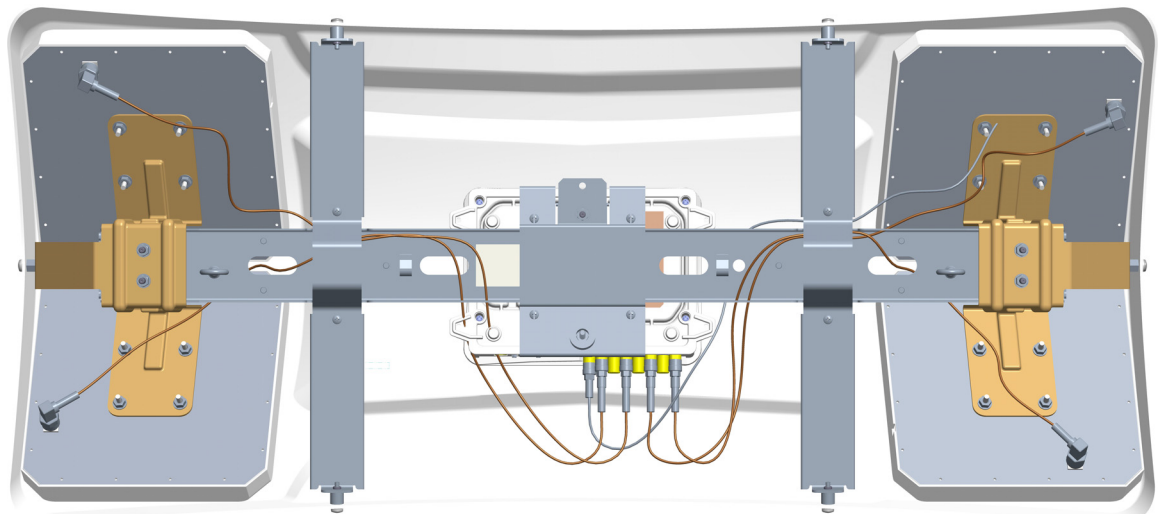
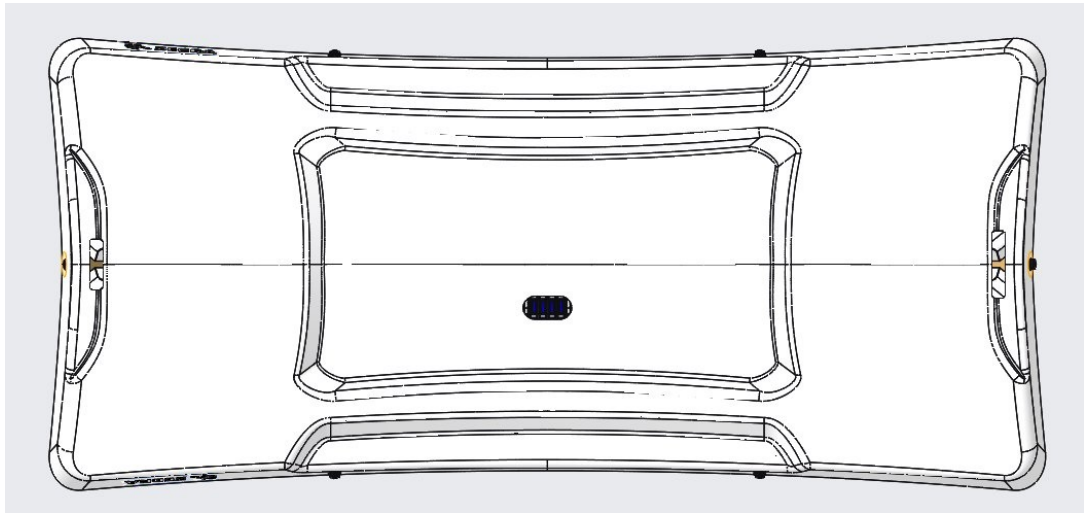


Figure 4 Bottom View



Mounting FX9600 Reader to ST5500

Introduction

This chapter provides instructions for mounting an FX9600 RFID reader (sold separately) to the ST5500.

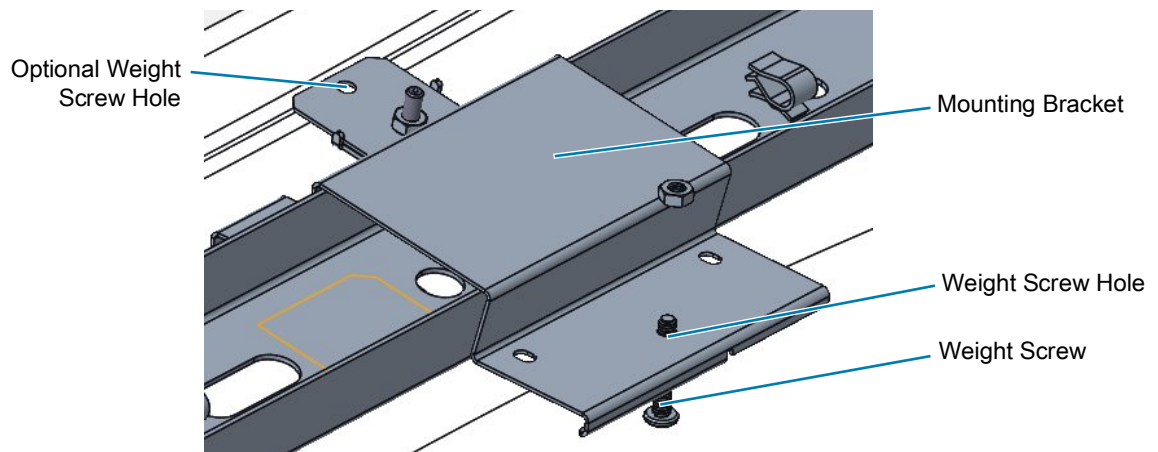
Mounting FX9600 to ST5500

Before mounting the FX9600 RFID reader to the ST5500, install the weight screws to the mounting bracket. Insert the weight screw to the weight screw hole as shown in [Figure 5](#) and tighten it with a nut (light torque or hand tighten).



NOTE: The mounting bracket has two weight screw holes (see [Figure 5](#)). Install the weight screws to these holes. The weights can be added on either weight screw as required when the final assembled unit is hung (see [Step 9 on page 22](#) for more details).

Figure 5 Install Weight Screw



Mounting FX9600 Reader to ST5500

To mount an FX9600 RFID reader to the ST5500 hanging transition reader:

1. Slide the FX9600 reader in place and hook/engage the rear of the reader into the bracket rear catch as shown in [Figure 6](#) and [Figure 7](#). Then, snap the front lip of the reader into the front catch of the bracket. The unit is suspended from the bracket.

Figure 6 Install the FX9600 Reader (Rear View)

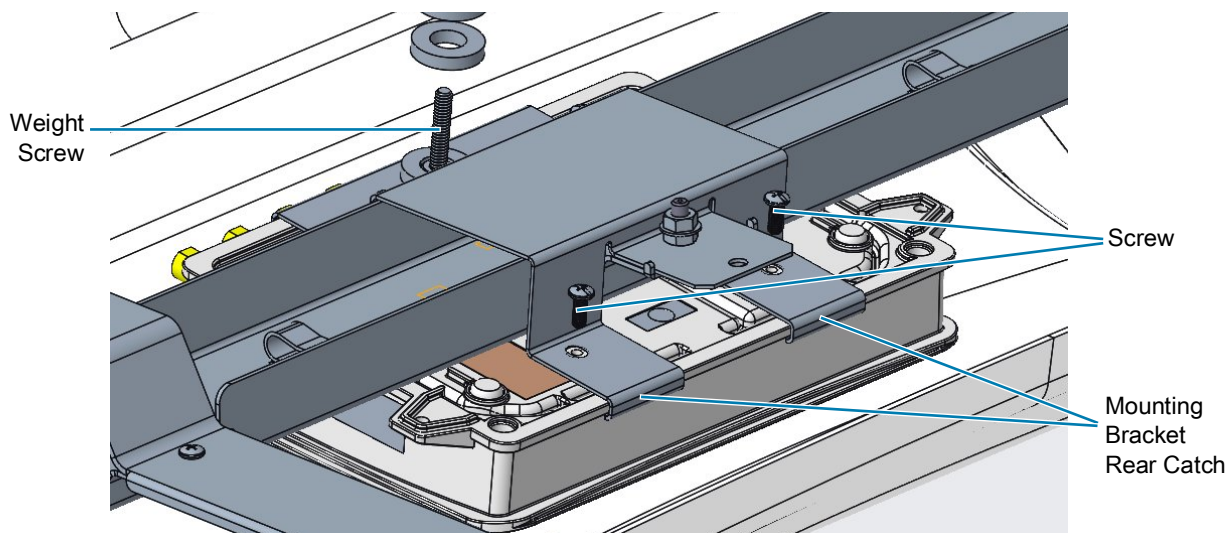
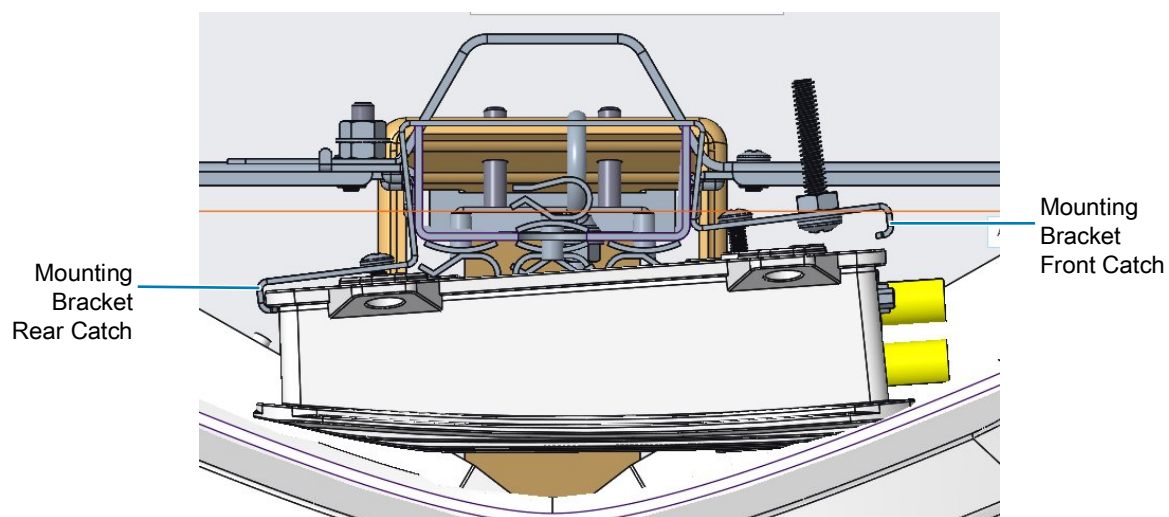


Figure 7 Mounted FX9600 Reader (Side View)

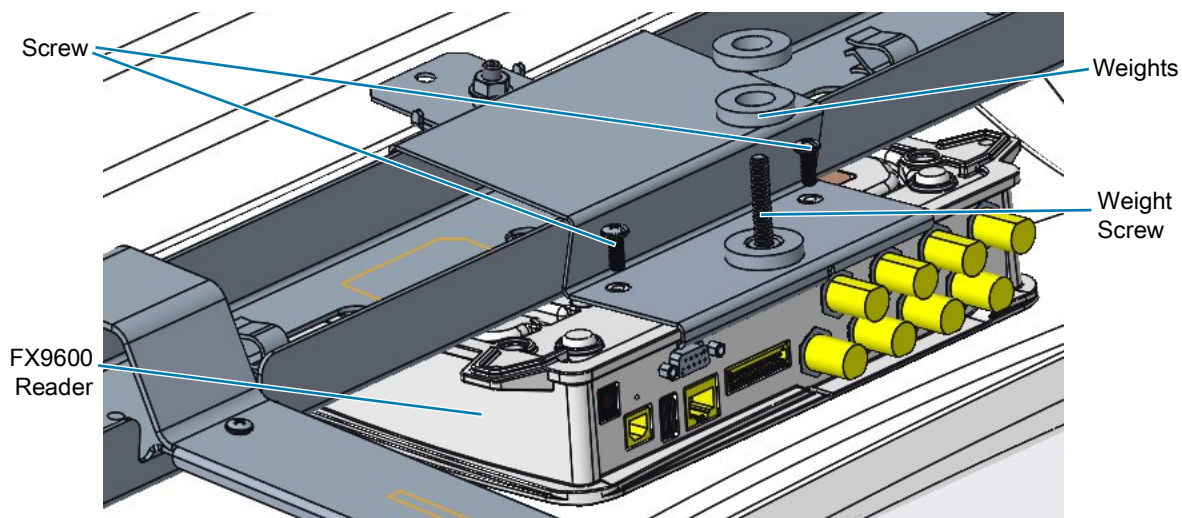


2. Screw the four provided screws with a hand screw driver or a tool with light torques (up to 10 in-lbs) as shown in [Figure 6 on page 12](#) and [Figure 8](#).



NOTE: The weights are added to either weight screw as required when the assembled unit is hung in the facility (see [Step 9 on page 22](#) for more details).

Figure 8 Install the FX9600 Reader (Front View)



3. Connect the four antenna cables on the ST5500 to the antenna ports on the FX9600.
4. To install the ST5500 in the facility, see [Installing ST5500 on page 14](#).

Installing ST5500

Introduction

Transition readers are located at the main transition points within a site. This chapter describes transition reader installation options and instructions.

Installation Options



IMPORTANT: Never install the ST5500 at a height greater than 12 ft measured from the finished floor to the underside of the unit.

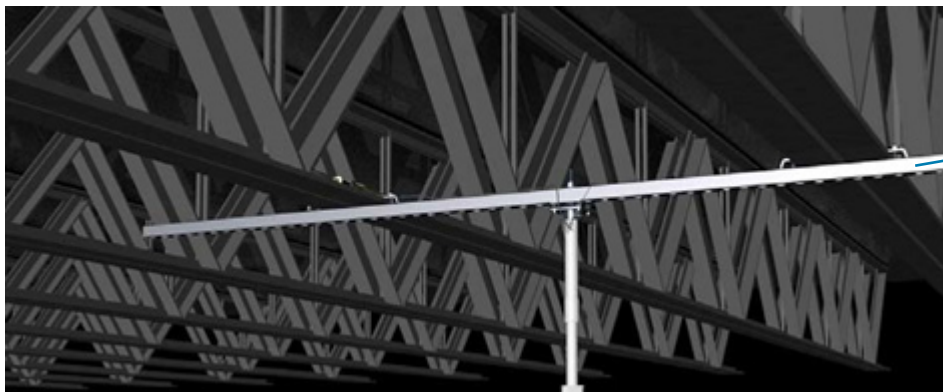
Users have various options to install the ST5500, depending on the site and the locations required. The options include: open ceiling mounted, drywall or drop ceiling mounted, wall mounted, and outdoor mounted. If there are no appropriate mounting features in the ceiling, select one of the following options for mounting the ST5500 that best suits the installation environment:

- [Installing Using a Strut Channel Clamped to the Bottom of the Truss on page 14](#)
- [Installing Using a Strut Channel and Threaded Rods Clamped to the Top of the Truss on page 15](#)

Installing Using a Strut Channel Clamped to the Bottom of the Truss

For locations where the ST5500 cannot be located directly under a truss, add a strut channel to the lower section of two trusses and mount the ST5500 on the strut channel.

Figure 9 Installing the ST5500 Using a Strut Channel



Strut
Channel

Secure the strut channel to two trusses using beam clamps appropriate for the installation. For example, insert the beam clamp (see [Figure 9](#) and [Figure 10](#)) into the strut channel and secure it to the beam with a bolt and channel nut.

Refer to the instructions provided with the beam clamp used.

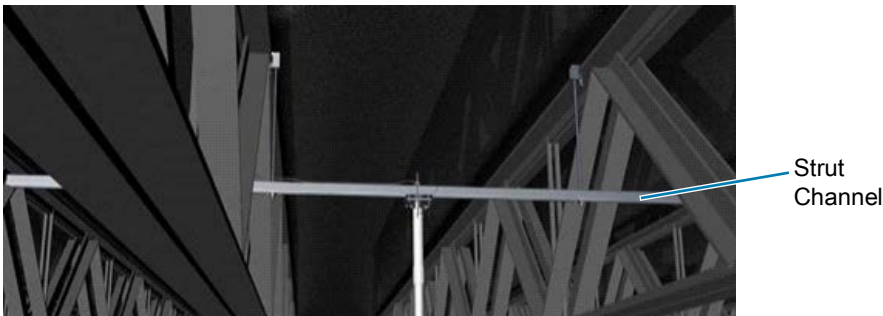
Figure 10 Sample Beam Clamps



Installing Using a Strut Channel and Threaded Rods Clamped to the Top of the Truss

For an alternative configuration using a strut channel, mount the strut channel to the top of the truss using threaded rods and beam clamps, and mount the ST5500 on the strut channel.

Figure 11 Installing the ST5500 to the Top of the Truss



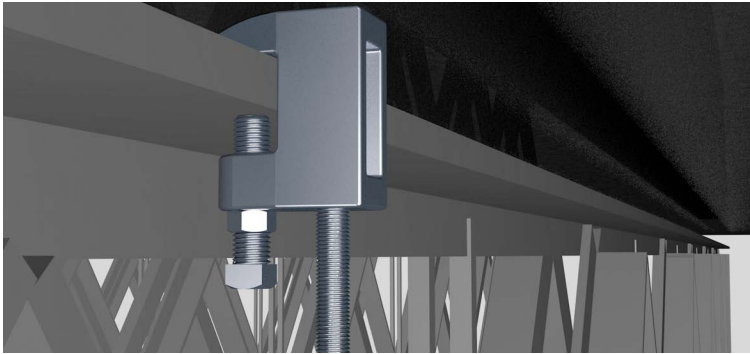
1. Secure a beam clamp appropriate for the installation to the top of each truss. Refer to the instructions provided with the beam clamp.

Figure 12 Sample Beam Clamp



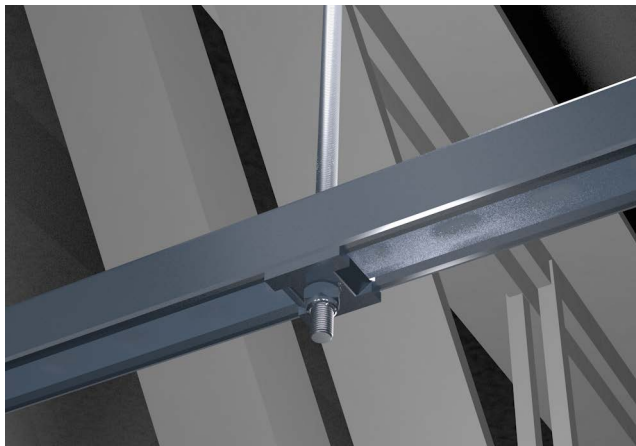
2. Insert a threaded rod into the bottom of each clamp.

Figure 13 Inserting a Threaded Rod Into the Beam Clamp



3. Use a flat washer and locknut from the top and a no-spin washer and locknut from the bottom to secure the threaded rods to each end of the strut channel.

Figure 14 Securing Threaded Rod to Strut Channel

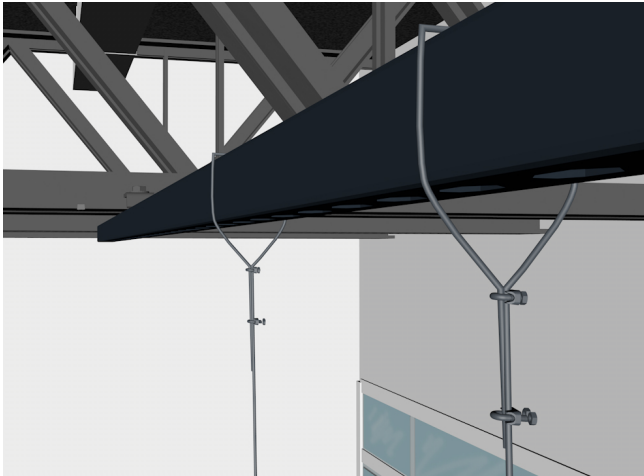


Installing the ST5500

To install an ST5500 Hanging Transition Reader:

1. Review the ceiling to identify mounting points above the designated ST5500 location. If there are no appropriate mounting features in this location, install a strut channel from which to hang the ST5500. See [Installation Options on page 14](#). Consider the following when selecting a location:
 - The ST5500 is typically installed near the site's entrance, centrally aligned with the doors and with the bottom of the unit just above the doorway.
 - Position the ST5500 at least 3 ft from the doorway(s), leaving an unobstructed view on both sides of the device.
2. Hang two pieces of wire rope (minimum diameter of 1/8 in. and a minimum load capacity of 200 lbs) from the ceiling infrastructure (truss or strut). Set the length of the wire rope so that the ST5500 is as low as possible without interfering with facility operations (such as above doorways and not blocking signs). Never install the ST5500 at a height greater than 12 ft measured from the finished floor to the underside of the unit.

Figure 15 Securing the Wire Rope



3. Loop the free ends of the wire rope, and secure them with two wire rope clamps to each end.

4. Attach a carabiner to each loop.



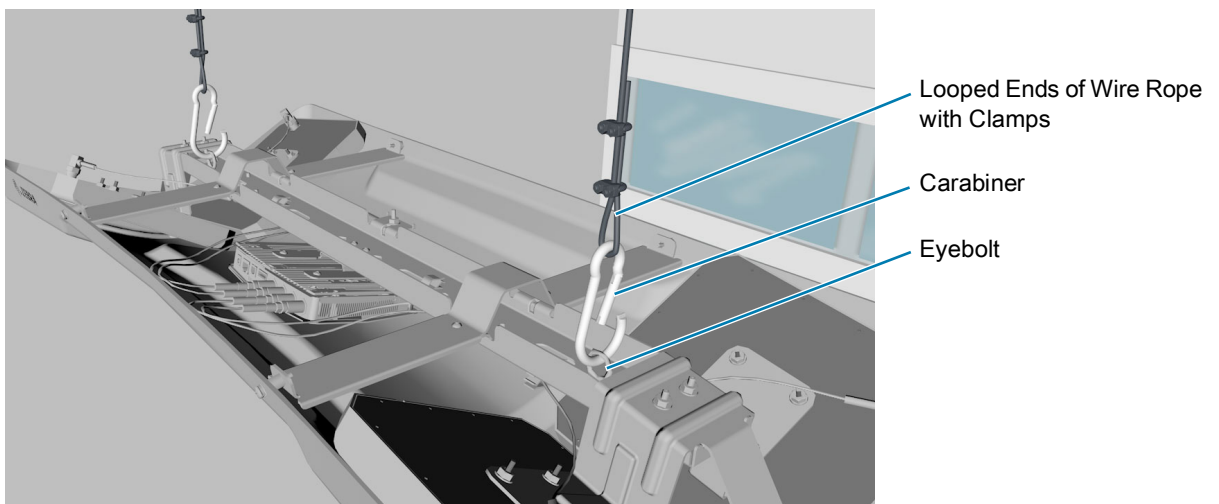
NOTE: The carabiners are located in a bag adhered to the inside flap of the ST5500 box.

5. Attach the carabiners to the eyebolts on the ST5500.



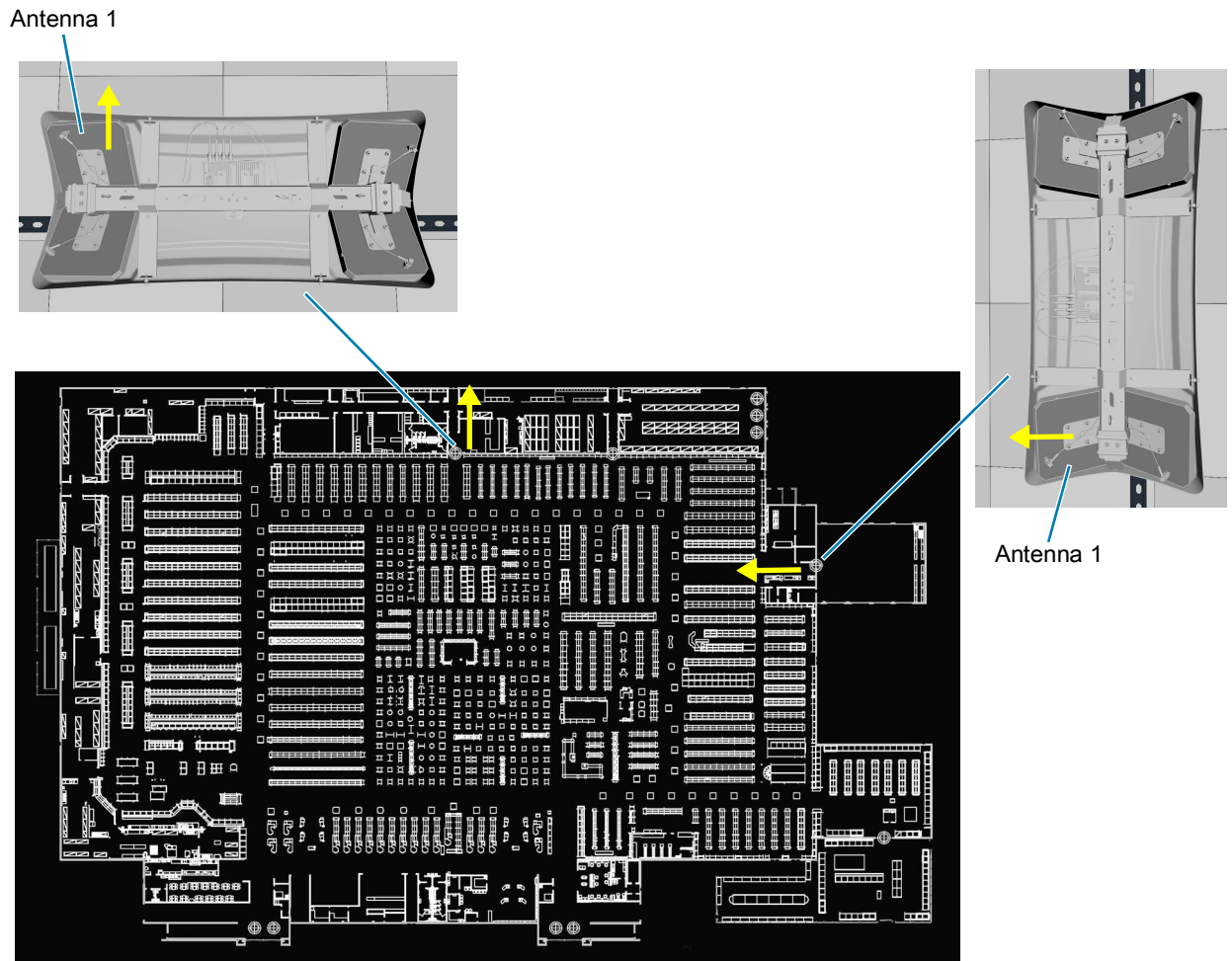
NOTE: The RFID reader illustrated in [Figure 16](#) may differ from the FX9600.

Figure 16 Attaching Carabiners to the Hanging Transition Reader



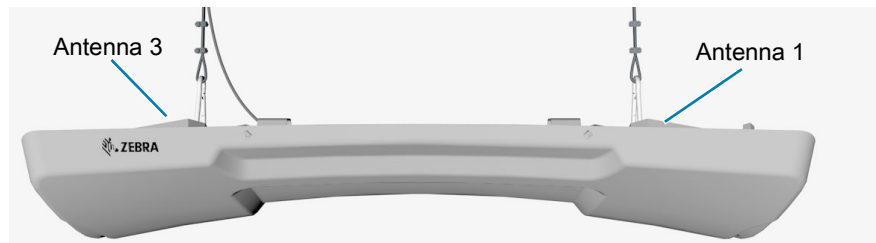
Orient the ST5500 so that antenna 1 (see [Figure 17](#)) always faces either the top or left side of the site map. Avoid physical obstructions in front of and below the reader.

Figure 17 Transition Reader Antenna 1 Location on the Map



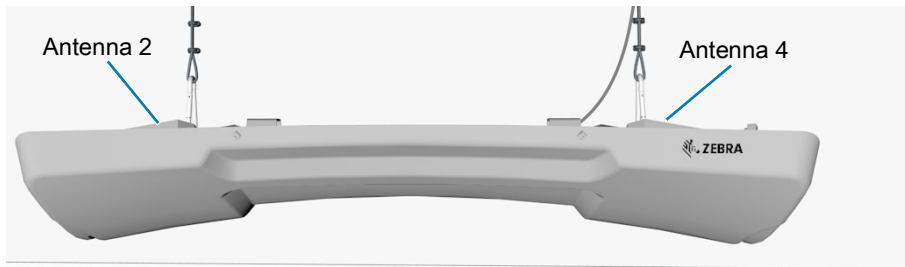
- When looking at the bottom of the ST5500 (see [Figure 18](#)), if the logo is on the left side, the antennas closest to the viewer are antenna 3 behind the logo on the left, and antenna 1 on the right.

Figure 18 Transition Reader Orientation 1



- If the logo is on the right side (see [Figure 19](#)), the antennas closest to the viewer are antenna 2 on the left, and antenna 4 behind the logo on the right.

Figure 19 Transition Reader Orientation 2

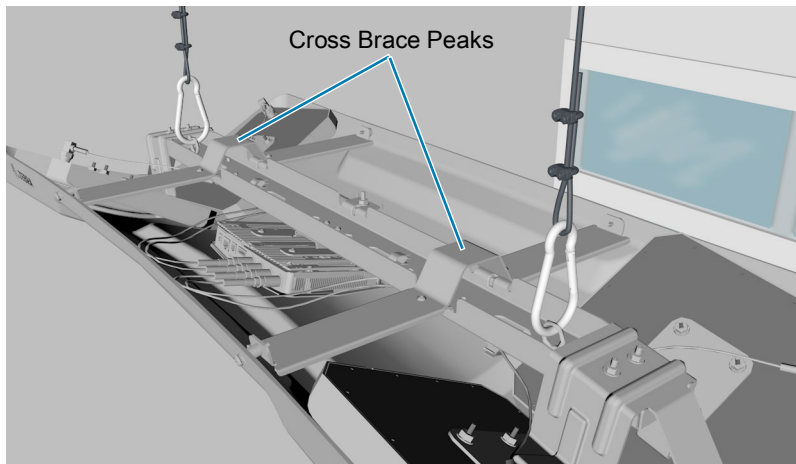


For drywall or drop ceiling installations where the ST5500 must be as close to the ceiling as possible, drill holes in the ceiling over the eyebolts for routing the wire rope, and bank the peaks of the two cross braces against the ceiling.



NOTE: The RFID reader illustrated in [Figure 20](#) may differ from the FX9600.

Figure 20 Cross Brace Peaks for Drop Ceilings



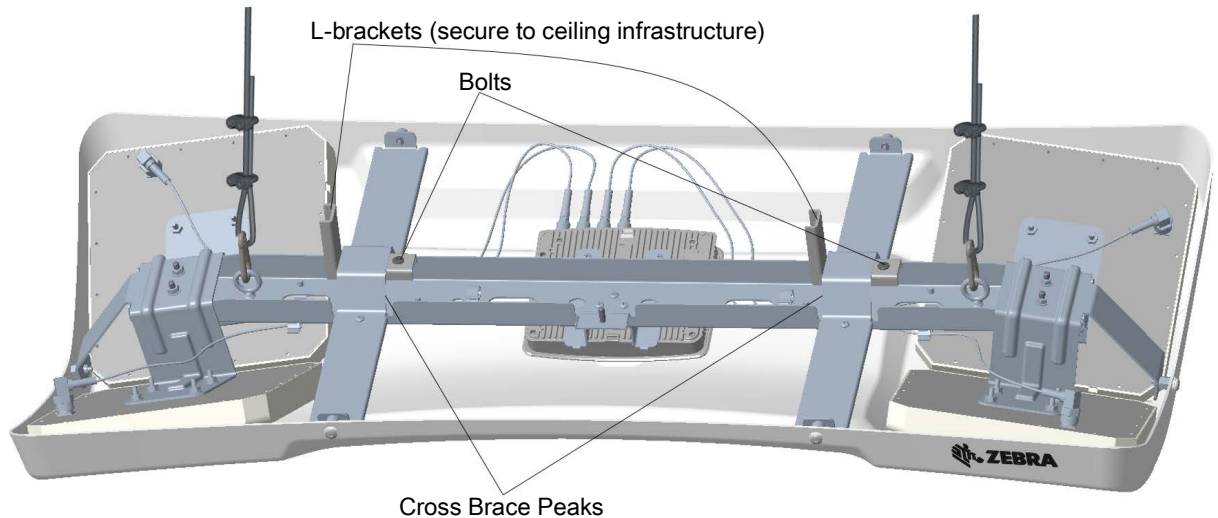
Optionally, two L-brackets or Z-brackets can be installed to the ceiling infrastructure and used as additional holding points for the ST5500 when the finished ceiling is low and the device needs to be banked against it. The L-brackets or Z-brackets do not replace the wire rope instructions detailed in [Step 2](#) but would instead allow the unit to first be suspended with the wire rope up to 6 in. from the finished ceiling and then easily slid onto the brackets for final positioning. This arrangement also makes it easier for servicing and removing the unit since the cables would have enough slack to drop the unit down after it is slid off the brackets, providing access to the carabiners. Ensure the following items are addressed:

- A small bolt and nut should be installed at the end of each L-bracket or Z-bracket, so the ST5500 cannot easily slide off the brackets. The unit will have to be lifted slightly to remove it.
- The horizontal section of the L-brackets or bottom leg of the Z-brackets should be offset from the finished ceiling just enough to provide room for the peaks of the ST5500's cross braces to slide over, leaving the ST5500 flush with the finished ceiling.
- All materials must be rated to support a load of at least 70 lbs.
- All work must follow industry best practices and conform to local codes.



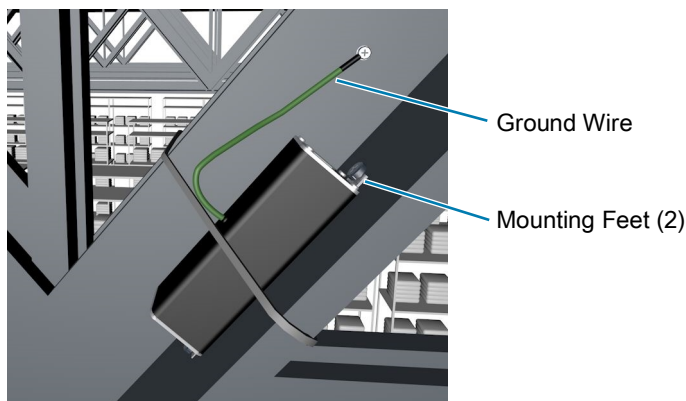
NOTE: The RFID reader illustrated in [Figure 21](#) may differ from the FX9600.

Figure 21 Optional Brackets (L-Brackets Shown)



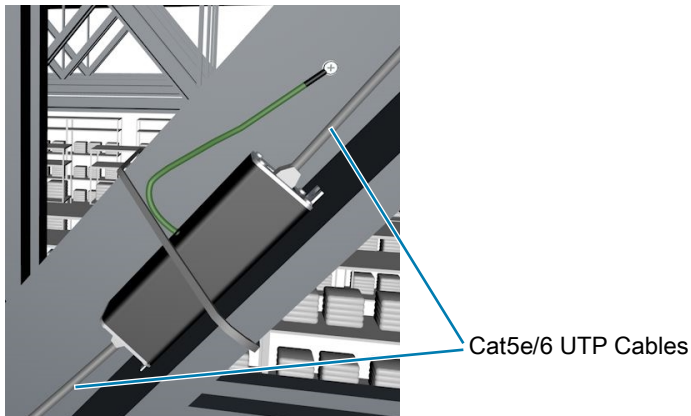
6. Level the ST5500 by adjusting the wire ropes as necessary.
7. Connect the Ethernet cable installed with the network infrastructure to the reader.
 - a. If using the optional surge protector, mount and ground it using one of the following methods:
 - Screwing the two mounting feet to the grounded metal building infrastructure.
 - Securing the surge protector to the truss and fastening the integral ground wire to the grounded metal building infrastructure.

Figure 22 Optional Surge Protector



- b. Plug the Cat5e/6 UTP cable from the midspan injector into one end of the surge protector.
- c. Plug Cat5e/6 UTP cable into the other end of the surge protector.

Figure 23 Connecting the Ethernet Cables to the Optional Surge Protector

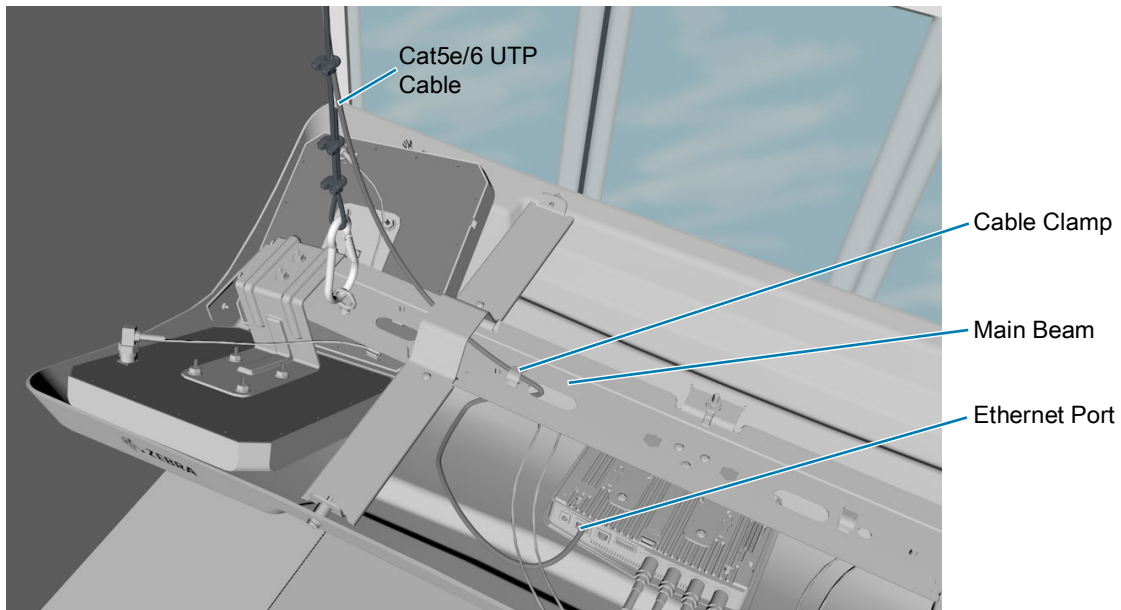


- d. Run the Cat5e/6 UTP cable alongside the wire rope, and cable-tie the two together as needed. Route the cable through the reader's main beam.
- e. Clip the cable onto the clamp on the main beam, and connect it to the Ethernet port on the reader.



NOTE: The RFID reader illustrated in [Figure 24](#) may differ from the FX9600.

Figure 24 Ethernet Cable Connection



NOTE: Ethernet surge protectors (Zebra p/n PWR-SGD00V30W-1NA) are optional but should be considered for locations known to be susceptible to lightning strikes or where Ethernet cables run from outdoor locations to indoor locations. These surge protectors offer increased protection for the products.

8. Measure the coordinates of the actual installed location of the ST5500 (relative to the appropriate site map datums), and submit them using the CMM application.

9. If the ST5500 is not leveled horizontally when it is hung, add the provided weights on either weight screw as required (see [Figure 25](#)) until it achieves final balance.

Figure 25 Add Weights on the Weight Screw

